# Diseases and immunity

Original exam-style questions · 20 questions

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## Question 1

core · 1 mark · 0610_T10_P1_20261013_Q1

Which statement describes a transmissible disease?

**A** a disease caused by a lack of one nutrient in the diet

**B** a disease in which the pathogen can be passed from one host to another

**C** a disease that is passed from parents to their offspring in their genes

**D** a disease that develops because body cells stop working properly with age

## Question 2

core · 1 mark · 0610_T10_P1_20261013_Q2

Which example shows a pathogen being transmitted indirectly?

**A** a nurse is infected when a patient's blood gets into a cut on her hand

**B** a child is infected after kissing a family member who has a cold sore

**C** a baby is infected by body fluids from its mother during birth

**D** an office worker touches a door handle used by an infected colleague, then touches his mouth

## Question 3

core · 1 mark · 0610_T10_P1_20261013_Q3

Which row correctly describes how mucus and stomach acid defend the body against pathogens?

| | **mucus** | **stomach acid** |
|---|---|---|
| A | traps pathogens | kills pathogens |
| B | kills pathogens | traps pathogens |
| C | traps pathogens | produces antibodies |
| D | produces antibodies | kills pathogens |

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## Question 4

supplement · 1 mark · 0610_T10_P1_20261013_Q4

Antibodies made against the measles virus do not bind to the chickenpox virus.

What is the reason for this?

**A** antibodies are only made against bacteria, not against viruses

**B** the chickenpox virus has no antigens on its surface

**C** the antibodies have a shape complementary to measles antigens but not to chickenpox antigens

**D** the antibodies are destroyed by phagocytes before they reach the chickenpox virus

## Question 5

supplement · 1 mark · 0610_T10_P1_20261013_Q5

Which statement describes antibodies?

**A** proteins that bind to antigens, leading to direct destruction of pathogens or marking of pathogens for destruction by phagocytes

**B** cells that engulf and digest pathogens

**C** carbohydrates that coat pathogens so that they cannot reproduce

**D** proteins that bind to any pathogen, whatever the shape of its antigens

## Question 6

supplement · 1 mark · 0610_T10_P1_20261013_Q6

What is put into the body during vaccination?

**A** antibodies made by another person

**B** memory cells

**C** white blood cells

**D** weakened pathogens or their antigens

## Question 7

supplement · 1 mark · 0610_T10_P1_20261013_Q7

Which row is correct for passive immunity?

| | **antibodies made by the person's own lymphocytes** | **memory cells produced** | **length of protection** |
|---|---|---|---|
| A | yes | yes | long-term |
| B | no | no | short-term |
| C | no | yes | long-term |
| D | yes | no | short-term |

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## Question 8

supplement · 1 mark · 0610_T10_P1_20261013_Q8

Which person has gained active immunity?

**A** a baby who receives antibodies in breast milk

**B** a fetus that receives antibodies across the placenta

**C** an adult who recovered from an infection and produced antibodies against the pathogen

**D** a patient who is injected with antibodies taken from the blood of another person

## Question 9

supplement · 1 mark · 0610_T10_P1_20261013_Q9

The list shows some events that happen after a person drinks water containing the cholera bacterium.

1 water moves into the gut by osmosis
2 the bacterium produces a toxin
3 chloride ions are secreted into the small intestine
4 diarrhoea causes dehydration

What is the correct order of these events?

**A** 2 → 1 → 3 → 4

**B** 3 → 2 → 1 → 4

**C** 2 → 3 → 4 → 1

**D** 2 → 3 → 1 → 4

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## Question 10

supplement · 1 mark · 0610_T10_P1_20261013_Q10

Which row describes cholera?

| | **type of pathogen** | **main way it is transmitted** |
|---|---|---|
| A | bacterium | in contaminated water |
| B | virus | in contaminated water |
| C | bacterium | in droplets in the air |
| D | fungus | in contaminated water |

## Question 11

core · 6 marks · 0610_T10_P1_20261013_Q11

After a large flood, the water pipes and sewers in a town were damaged. Rubbish was not collected for several weeks. Many people then became ill with a disease that causes diarrhoea. The pathogen was found in the drinking water and in piles of uncollected rubbish.

(a) The disease is a transmissible disease. State what is meant by a transmissible disease. [1]

(b) Drinking contaminated water is one indirect way in which a pathogen can be transmitted. State two other indirect ways in which pathogens can be transmitted. [2]

(c) Explain why repairing the sewage treatment system and collecting the rubbish would help to control the spread of the disease. [3]

## Question 12

core · 5 marks · 0610_T10_P1_20261013_Q12

The human body has several defences against pathogens.

(a) Describe how hairs in the nose and mucus help to stop pathogens in the air from entering the lungs. [2]

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(b) Explain why a deep cut in the skin increases the risk of infection. [1]

(c) Some pathogens get past these defences and enter the blood. Describe how white blood cells defend the body against these pathogens. [2]

## Question 13

core · 7 marks · 0610_T10_P1_20261013_Q13

The manager of a school canteen recorded the number of students with food poisoning in two terms.

At the start of term 2, two new rules were introduced for canteen staff:

- rule 1: wash hands with soap after using the toilet and before handling food
- rule 2: store and prepare raw meat separately from cooked food

The table shows the results.

| **term** | **number of students eating in the canteen** | **number of students with food poisoning** |
|---|---|---|
| term 1 (before the rules) | 1200 | 48 |
| term 2 (after the rules) | 1250 | 15 |

(a) Calculate the percentage of students eating in the canteen who had food poisoning in each term. [2]

(b) Explain why it is better to compare these percentages than to compare the numbers of students with food poisoning. [1]

(c) Explain how each of the two rules helps to reduce food poisoning. [4]

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## Question 14

supplement · 7 marks · 0610_T10_P1_20261013_Q14

A person was given a first dose of a vaccine on day 0 and a second dose of the same vaccine on day 60. The table shows the concentration of antibodies against the pathogen in the person's blood.

| time after first dose / days | antibody concentration / arbitrary units |
|---|---|
| 0 | 0 |
| 10 | 15 |
| 20 | 30 |
| 30 | 20 |
| 40 | 10 |
| 60 | 5 |
| 70 | 160 |
| 80 | 240 |
| 100 | 200 |

(a) State what is put into the body in a vaccine. [1]

(b) Outline how the first dose of the vaccine leads to an increase in antibody concentration between day 0 and day 20. [2]

(c) Use the data to compare the response to the second dose with the response to the first dose. [2]

(d) Explain why the response to the second dose is different from the response to the first dose. [2]

## Question 15

supplement · 6 marks · 0610_T10_P1_20261013_Q15

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The table shows the percentage of children vaccinated against measles and the number of measles cases in one country over 15 years.

| year | percentage of children vaccinated | number of measles cases |
|---|---|---|
| 2000 | 45 | 24 000 |
| 2005 | 60 | 15 600 |
| 2010 | 80 | 4800 |
| 2015 | 95 | 360 |

(a) Describe the relationship between the percentage of children vaccinated and the number of measles cases. [1]

(b) Calculate the percentage decrease in the number of measles cases between 2000 and 2015. [2]

(c) Explain how vaccinating a large percentage of the population reduces the spread of measles, including to people who have not been vaccinated. [3]

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## Question 16

supplement · 6 marks · 0610_T10_P1_20261013_Q16

Newborn babies are protected against some pathogens by antibodies from their mother.

(a) State how antibodies pass from the mother to the fetus before birth. [1]

(b) Explain the importance of breast-feeding for the development of passive immunity in a young baby. [3]

(c) The protection given by passive immunity lasts only a short time. Explain why. [2]

## Question 17

supplement · 7 marks · 0610_T10_P1_20261013_Q17

Cholera is a disease that causes severe diarrhoea.

(a) State the type of organism that causes cholera and how cholera is transmitted. [2]

(b) Explain how the cholera pathogen causes diarrhoea. [3]

(c) The cholera pathogen stays inside the gut. Explain why a person with cholera can still become dangerously ill. [2]

## Question 18

supplement · 5 marks · 0610_T10_P1_20261013_Q18

Two strains of a bacterium, strain X and strain Y, cause the same type of illness. Strain X and strain Y have different antigens.

One woman recovered from an infection by strain X. One year later she was infected by strain Y and became ill again.

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(a) State how the antigens of strain X differ from the antigens of strain Y. [1]

(b) Explain why the antibodies that the woman made against strain X did not protect her against strain Y. [2]

(c) Describe two ways in which antibodies that bind to antigens lead to the destruction of a pathogen. [2]

## Question 19

supplement · 6 marks · 0610_T10_P1_20261013_Q19

Two scientists are going to work for one year in a region where a viral disease, disease P, is common.

- Scientist 1 is vaccinated against disease P eight weeks before travelling.
- Scientist 2 has no time to be vaccinated. The day before travelling, she is injected with antibodies against the disease P virus that were collected from the blood of another person.

(a) State the type of immunity each scientist has against disease P when they arrive. [2]

(b) Predict which scientist is more likely to still be protected against disease P at the end of the year. Explain your answer. [4]

## Question 20

supplement · 7 marks · 0610_T10_P1_20261013_Q20

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There was an outbreak of cholera in a city. The city has two districts:

- district K gets its drinking water from a treated, piped supply, and its sewage is treated
- district L gets its drinking water from a river, and untreated sewage flows into the river upstream of the district

The table shows the number of cholera cases in each district during the outbreak.

| district | population | number of cholera cases |
|---|---|---|
| K | 40 000 | 60 |
| L | 25 000 | 900 |

(a) Calculate the number of cholera cases per 1000 people in each district. [2]

(b) Use the information to explain the difference in the number of cases per 1000 people in the two districts. [3]

(c) Health officials plan to vaccinate people in district L against cholera. Suggest why they should also provide district L with a clean water supply. [2]

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